

Evaluating and Equivalent Expressions

Teacher's Edition

The Period

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

Must Integer exponents and rational bases, with procedural fluency: evaluate, and decide equivalence by rewriting to a common base.

Must not Bases that get rewritten are powers of 2, 3 or 5. No prime factorization of arbitrary numbers, and no fractional exponents.

Target I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

MTR MTR.2.1 — Notes I · **MTR.5.1** — Notes III and boards 4, 7 and 8 · **MTR.6.1** — Board 9

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-7	14 min
Example 1	8-9	3 min
Your Turn	10-11	3 min
Example 2	12-13	3 min
Your Turn	14-15	3 min
Whiteboards	16-33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		53 min

Say these three things out loud today

1. Different bases cannot be compared. Rewrite both to the same base first, then look.
2. Every law you know can appear in one expression at once. The order is yours; the value is not.
3. Test a claim before you agree with it. Convert, then decide.

Slide by Slide

1. **Evaluating and Equivalent Expressions.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite for the second half of today: one number, three bases.
3. **Warm-Up — Answers.** Reveal. 1. $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$ 2. 25 3. 100 4. 2^6 , 4^3 , 8^2
4. **Notes.** Melanie applies the product law first ($-2 + 5 = 3$), then the quotient law.
5. **Notes — Copy the worked line.** Power of a power ($2 \cdot -3 = -6$), then quotient ($-6 - (-5) = -1$), then the negative exponent law to finish.
6. **Notes.** 8 is 2^3 , so 8^5 is $(2^3)^5 = 2^{15}$.
7. **Notes.** Xavier says both expressions rewrite to 2^{-7} .
8. **Example 1 — Evaluate.** SPOKEN ONLY.
9. **Example 1 — Worked.** Flip, square, then multiply by 3.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a student who folds the outside 2 into the base (a board with $(4/5)^{-2}$, or 25/16) and one who never flipped ($4/25 \cdot 2 = 8/25$).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Equivalent or not?** SPOKEN ONLY.
13. **Example 2 — Worked.** Left: $6^3 = (3 \cdot 2)^3 = 3^3 \cdot 2^3$, so $3^5 \cdot 2^3$.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a student who cannot split 10 into $2 \cdot 5$ (board stops at 10^3) and one who subtracts the wrong exponent on the right.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 16-17. **Board 1.** $\frac{(-7)^5}{(-7)^0 \cdot (-7)^2} \rightarrow \textbf{-343}$ 343 — sign dropped on an odd power; 'undefined' — $(-7)^0$ in the denominator read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
- 18-19. **Board 2.** $\frac{(\frac{3}{4})^{-3} \cdot (\frac{3}{4})^0}{(\frac{3}{4})^{-2}} \rightarrow \frac{4}{3}$ 3/4 — $(3/4)^{-1}$ left unflipped [negative exponent law]; $(3/4)^{-5}$ or $(4/3)^5$ — subtracted $-3 - 2$ [integer subtraction of a negative]
- 20-21. **Board 3.** $\frac{(2^{-1})^3 \cdot (\frac{1}{2})^{-4}}{2 \cdot 2^2} \rightarrow \frac{1}{4}$ 4 — 2^{-2} left unflipped [negative exponent law]; $1/2$ — the denominator $2 \cdot 2^2$ counted as 2^2 [product of powers]
- 22-23. **Board 4.** One flash drive holds 2^{15} megabytes. Another holds 8^5 megabytes. The first one costs 45 dollars. Which holds more — or is it the same? Show why. $\rightarrow$ **The same: $8^5 = (2^3)^5 = 2^{15}$** The price of 45 dollars is not needed. ' 8^5 holds more' — the bases were compared, not the values [8.NSO.1.3: rewrite to a common base]; ' 2^{15} holds more' — the exponents were compared; 2^8 — added $3 + 5$ in the power-of-a-power step [confused with product of powers]; anything built from 45 — the unneeded figure was used
- 24-25. **Board 5.** 9^4 Which expression is equivalent? $\rightarrow \textbf{C - } 3^8$ (A) squared the exponent, $4^2 = 16$ [the outer exponent applied to the exponent]; (B) replaced 9 by 3 and kept the exponent [the 3^2 was not written]; (D) added $2 + 4$ instead of multiplying [power

of a power confused with product of powers] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

26-27. Board 6. $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ Which expression is equivalent? → **A — 2^{-7}** (B) dropped both minus signs [negative exponent law: a negative exponent does not disappear]; (C) wrote $1/4$ as 2^{-1} instead of 2^{-2} [the base 4 is 2^2 , not 2]; (D) multiplied $-4 \cdot 3$ [confused with power of a power] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

28-29. Board 7. $2^3 \cdot 4^3$ → **2^9** 8^3 or 8^6 — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]; $2^6 - 4^3$ rewritten but the 2^3 dropped

30-31. Board 8. $4^7 \cdot 4^{-2}$ → **2^{10}** $2^5 - 4$ replaced by 2 without the exponent 2; 4^9 or 2^{18} — subtracted instead of adding a negative

32-33. Board 9. Xavier says $(1/4)^2 \cdot 2^{-3}$ and $(1/2)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-7} . Do you agree? → **No — the second is 2^{-6} , not 2^{-7}** 'Yes' — 4^{-1} rewritten as 2^{-1} [the base 4 is 2^2]

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 343 — sign dropped on an odd power

Board 2. $3/4 - (3/4)^{-1}$ left unflipped [negative exponent law]

Board 3. $4 - 2^{-2}$ left unflipped [negative exponent law]

Board 4. ' 8^5 holds more' — the bases were compared, not the values [8.NSO.1.3: rewrite to a common base]

Board 5. squared the exponent, $4^2 = 16$ [the outer exponent applied to the exponent]

Board 6. dropped both minus signs [negative exponent law: a negative exponent does not disappear]

Board 7. 8^3 or 8^6 — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]

Board 8. $2^5 - 4$ replaced by 2 without the exponent 2

Board 9. 'Yes' — 4^{-1} rewritten as 2^{-1} [the base 4 is 2^2]

Materials Whiteboards and markers. Calculators are allowed on everything and are how students confirm $8^5 = 2^{15} = 32,768$ once.